

Practical Workbook

CT-261

DATABASE MANAGEMENT SYSTEMS



Name: **Moazzam Farooqui**

Roll No: **CT-24068**

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Department: **CSIT**

Practical Workbook

Database Management Systems

(CT - 261)

Prepared by
Dr. Umer Farooq
CS & IT

Approved by
Chairman

Department of Computer Science & Information Technology NED University of
Engineering & Technology

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LAB#10

Creating Views

Note: HR dataset was downloaded and lab was performed on that particular dataset so names of the table have been adjusted accordingly (e.g. OEHR_EMPLOYEES, OEHR_DEPARTMENTS)

Q1. Create a view called EMPLOYEES_VU based on the employee numbers, employee names, and department numbers from the EMPLOYEES table. Change the heading for the employee name to EMPLOYEE.

SOLUTION:

```
1  -- Q1
2  Create View EMPLOYEES_VU As
3  Select EMPLOYEE_ID, LAST_NAME As "EMPLOYEE", DEPARTMENT_ID
4  From OEHR_EMPLOYEES;
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
View created.				
0.03 seconds				

Q2. Display the contents of the EMPLOYEES_VU view.

SOLUTION:

```
6  -- Q2
7  Select * From EMPLOYEES_VU;
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
EMPLOYEE_ID	EMPLOYEE	DEPARTMENT_ID		
100	King	90		
101	Kochhar	90		
102	De Haan	90		
103	Hunold	60		
104	Ernst	60		
105	Austin	60		

Q3. Select the view name and text from the USER_VIEWS data dictionary view.

Note: Another view already exists. The EMP_DETAILS_VIEW was created as part of your schema.

Note: To see more contents of a LONG column, use the iSQL*Plus command SET LONG n, where n is the value of the number of characters of the LONG column that you want to see.

SOLUTION:

```
9  -- Q3
10 -- This command cannot be used in Oracle APEX
11 Set Long 2000;
```

```
13 Select VIEW_NAME, TEXT
14 From USER_VIEWS;
```

OUTPUT:

VIEW_NAME	TEXT
DEPT50	SELECT EMPNO,ENAME AS "EMPLOYEE",DEPTNO FROM EMP WHERE DEPTNO=50 WITH CHECK OPTION
EMPLOYEES_VU	Select EMPLOYEE_ID, LAST_NAME As "EMPLOYEE", DEPARTMENT_ID From OEHR_EMPLOYEES
EMP_DEPT_V	select e.empno, e.ename, e.job, (select m.ename from emp m where e.mgr = m.empno) mgr, e.hiredate, e.sal, e.comm, d.deptno, d.dname, d.loc from emp e, dept d where e.deptno = d.deptno (+)
OEHR_ACCOUNT_MANAGERS	select c.account_mgr_id acct_mgr, cr.region_id region, c.country_id country, c.state_province province, count(*) num_customers from oeher_customers c, oeher_countries cr where c.country_id = cr.country_id group by rollup (c.account_mgr_id , cr.region_id , c.country_id , c.state_province)
OEHR_BOMBAY_INVENTORY	select p.product_id, p.product_name, i.quantity_on_hand from oeher_inventories i, oeher_warehouses w, oeher_products p where p.product_id = i.product_id and i.warehouse_id = w.warehouse_id and w.warehouse_name = 'Bombay'
OEHR_CUSTOMERS_VIEW	select customer_id, c.cust_first_name, c.cust_last_name, c.street_address, c.postal_code, c.city, c.state_province, co.country_id, co.country_name, co.region_id, c.nls_language, c.nls_territory, c.credit_limit, c.cust_email, c.phone_number, c.account_mgr_id from oeher_countries co, oeher_customers c where c.country_id = co.country_id(+)
OEHR_EMP_DETAILS_VIEW	select e.employee_id, e.job_id, e.manager_id, e.department_id, d.location_id, l.country_id, e.first_name, e.last_name, e.salary, e.commission_pct, d.department_name, j.job_title, l.city, l.state_province, c.country_name, r.region_name from oeher_employees e, oeher_departments d, oeher_jobs j, oeher_locations l, oeher_countries c, oeher_regions r where e.department_id = d.department_id and d.location_id = l.location_id and l.country_id = c.country_id and c.region_id = r.region_id and j.job_id = e.job_id with read only
OEHR_ORDERS_VIEW	select order_id, to_date(to_char(order_date, 'DD-MON-YY HH:MM:SS'), 'DD-MON-YY HH:MM:SS') order_date, order_mode, customer_id, order_status, order_total, sales_rep_id, promotion_id from oeher_orders
	select i.product_id, d.language_id, case when d.language_id is not null then d.translated_name else translate(i.product_name using nchar_cs) end as product_name, i.category_id, case when d.language_id is not null then d.translated_description else translate(i.product_description using nchar_cs) end as product_description, i.weight_class, i.warranty_period, i.supplier_id, i.product_status, i.list_price, i.min_price, i.catalog_url from oeher_product_information i, oeher_product_descriptions d where d.product_id (+) = i.product_id and d.language_id (+) = sys_context('USERENV', 'LANG')
OEHR_PRODUCT_PRICES	select category_id, count(*) as "#_OF_PRODUCTS", min(list_price) as low_price, max(list_price) as high_price from oeher_product_information group by category_id
More than 10 rows available. Increase rows selector to view more rows.	

Q4. Using your EMPLOYEES_VU view, enter a query to display all employee names and department numbers.

SOLUTION:

```
17  -- Q4
18  Select EMPLOYEE,DEPARTMENT_ID
19  From EMPLOYEES_VU;
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
EMPLOYEE		DEPARTMENT_ID		
King		90		
Kochhar		90		
De Haan		90		
Hunold		60		
Ernst		60		
Austin		60		

Q5. Create a view named DEPT50 that contains the employee numbers, employee last names, and department numbers for all employees in department 50. Label the view columns EMPNO, EMPLOYEE, and DEPTNO. Do not allow an employee to be reassigned to another department through the view.

SOLUTION:

```
21  -- Q5
22  Create View DEPT50 As
23  Select EMPLOYEE_ID As "EMPNO",LAST_NAME As "EMPLOYEE",DEPARTMENT_ID As "DEPTNO"
24  From OEHR_EMPLOYEES
25  Where DEPARTMENT_ID=50
26  With Check Option;
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
View created.				
0.03 seconds				

Q6. Display the structure and contents of the DEPT50 view.

SOLUTION:

```
28  -- Q6
29  -- To see the structure
30  Desc DEPT50;
```

OUTPUT:

Results Explain Describe Saved SQL History									
Object Type		VIEW ?							
		Object DEPT50 ?							
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DEPT50	EMPNO	NUMBER	-	6	0	-	-	-	-
	EMPLOYEE	VARCHAR2	25	-	-	-	-	-	-
	DEPTNO	NUMBER	-	4	0	-	✓	-	-

SOLUTION:

```
32  -- To see the data
33  Select * From DEPT50;
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
EMPNO		EMPLOYEE		DEPTNO
120		Weiss		50
121		Fripp		50
122		Kaufing		50
123		Vollman		50
124		Mourgos		50
125		Nayer		50

Q7. Attempt to reassign Matos to department 80.

SOLUTION:

```
35  -- Q7
36  Update DEPT50
37  Set DEPTNO=80
38  Where EMPLOYEE='Matos';
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
ORA-01402: view WITH CHECK OPTION where-clause violation				
0.03 seconds				

Q8. Create a view called SALARY_VU based on the employee last names, department names, salaries, and salary grades for all employees. Use the EMPLOYEES, DEPARTMENTS, and JOB_GRADES tables. Label the columns Employee, Department, Salary, and Grade, respectively.

SOLUTION:

```
40  -- Q8
41  ✓ Create Table JOB_GRADES(
42      GRADE_LEVEL Varchar2(5),
43      LOWEST_SALARY Number,
44      HIGHEST_SALARY Number
45  );
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
Table created.				
0.03 seconds				

SOLUTION:

```
47 Insert Into JOB_GRADES Values('A',1000,1999);
48 Insert Into JOB_GRADES Values('B',2000,2999);
49 Insert Into JOB_GRADES Values('C',3000,3999);
50 Insert Into JOB_GRADES Values('D',4000,4999);
51 Insert Into JOB_GRADES Values('E',5000,5999);
52 Insert Into JOB_GRADES Values('F',6000,30000);
```

```
54 Create View SALARY_VU As
55 Select E.LAST_NAME As "Employee",D.DEPARTMENT_NAME As "Department",E.SALARY As "Salary",G.GRADE_LEVEL As "Grade"
56 From OEHR_EMPLOYEES E
57 Join OEHR_DEPARTMENTS D On E.DEPARTMENT_ID=D.DEPARTMENT_ID
58 Join JOB_GRADES G On E.SALARY Between G.LOWEST_SALARY And G.HIGHEST_SALARY;
```

OUTPUT:

```
60  Select * From SALARY_VU
```

Results

Explain

Describe

Saved SQL

History

Employee	Department	Salary	Grade
Whalen	Administration	4400	D
Fay	Marketing	6000	F
Hartstein	Marketing	13000	F
Colmenares	Purchasing	2500	B
Himuro	Purchasing	2600	B
Tobias	Purchasing	2800	B
Baida	Purchasing	2900	B
Khoo	Purchasing	3100	C
Raphaely	Purchasing	11000	F
Mavris	Human Resources	6500	F

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.20 seconds

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